$P = W$

Nir Elber

Fall 2026

Contents

Contents 1

1 September 21: Overview of $P = W$ 1
- 1.1 Hodge Theory . . . 1
- 1.2 The Riemann–Hilbert Correspondence . . . 2
- 1.3 Nonabelian Hodge Theory . . . 3
- 1.4 The Statement . . . 4

2 September 28: Hitchin Fibrations and Spectral Curves 5
- 2.1 The Hitchin Fibration . . . 5
- 2.2 Spectral Curves . . . 6
- 2.3 Properness . . . 7

3 October 19: Mixed Hodge Structures 8
- 3.1 Pure Hodge Structures . . . 8
- 3.2 Mixed Hodge Structures . . . 9
- 3.3 Mixed Hodge Complexes . . . 10
- 3.4 Simplicial Varieties . . . 11

4 October 26: Perverse Sheaves and Perverse Filtrations 12
- 4.1 Constructible Sheaves . . . 12
- 4.2 Perverse Sheaves . . . 13
- 4.3 Simple Perverse Sheaves . . . 14
- 4.4 The Perverse Filtration . . . 15

1 September 21: Overview of $P = W$

This talk was givenby Leonid at MIT for the $P = W$ seminar.

1.1 Hodge Theory

Our goal today is to state the $P = W$ conjecture. Our main task will be to say something about nonabelian Hodge theory. Let's start with usual Hodge theory.

Theorem 1 (de Rham)**.** Fix a smooth projective variety X over $\mathbb{C}$. Then there is a natural isomorphism

$$\mathrm{H}^\bullet_{\mathrm{dR}}(X/\mathbb{C}) \to \mathrm{H}^\bullet_{\mathrm{B}}(X;\mathbb{C}).$$

Sketch. This is a long and difficult theorem. The map sends a differential k-form $[\alpha] \in \mathrm{H}^k_{\mathrm{dR}}(X/\mathbb{C})$ to the smooth singular cocycle

$$\gamma \mapsto \int_\gamma \alpha.$$

We will not run any of the checks. ■

Notation 2 (Dolbeault cohomology)**.** Fix a smooth projective variety X over $\mathbb{C}$. Then we define

$$\mathrm{H}^{p,q}(X) \coloneqq \mathrm{H}^q(X; \Omega^p_X).$$

This can be realized as the H^q-cohomology of a certain fine resolution $\{\mathcal{A}^{p,\bullet}(X)\}$ of sheaves, where $\mathcal{A}^{p,q}(X)$ is the space of (p,q)-differential forms.

Theorem 3. Fix a smooth projective variety X over $\mathbb{C}$. Then there is a canonical isomorphism

$$\mathrm{H}^\bullet_{\mathrm{dR}}(X/\mathbb{C}) \cong \bigoplus_{p,q} \mathrm{H}^{p,q}(X).$$

Sketch. Start by choosing a Hermitian metric on X, which gives rise to an elliptic differential Laplacian operator Δ. Then define $\mathcal{H}^k(X)$ to be the kernrel of Δ on $\mathcal{A}^k(X)$, and define $\mathcal{H}^{p,q}(X)$ similarly. It turns out that $\mathrm{H}^k_{\mathrm{dR}}(X)$ is isomorphic to $\mathcal{H}^k(X)$, and $\mathrm{H}^{p,q}(X)$ is isomorphic to $\mathcal{H}^{p,q}(X)$, and the isomorphism follows by decomposing a differential form into holomorphic and anti-holomorphic parts. ■

1.2 The Riemann–Hilbert Correspondence

We now pass to nonabelian Hodge theory. The goal is to retell the story with non-constant coefficients. Let's start with the Betti side.

Definition 4. Fix a smooth projective curve C of genus $g \geq 1$. Then we define the *Betti moduli space* $\mathcal{M}_{n,\mathrm{B}}$ to be the space of semisimple representations of $\pi_1(C(\mathbb{C}))$ of dimension n.

Remark 5. One can present $\pi_1(C(\mathbb{C}))$ as $\langle \alpha_1, \dots, \alpha_g, \beta_1, \dots, \beta_g : [\alpha_1, \beta_1] \cdots [\alpha_g, \beta_g] = 1 \rangle$. Thus, $\mathcal{M}_{n,\mathrm{B}}$ can be realized as

$$\left\{ (A_1, \dots, A_g, B_1, \dots, B_g) \in \mathrm{GL}_n(\mathbb{C}) : \prod_{i=1}^g [A_i, B_i] = 1 \right\} / \mathrm{GL}_n(\mathbb{C}).$$

(This quotient is a GIT quotient.) Thus, $\mathcal{M}_{n,\mathrm{B}}$ is an affine algebraic variety.

Example 6. If $n = 1$, then $\mathcal{M}_{1,\mathrm{B}}$ is $(\mathbb{C}^\times)^{2g}$.

Here is the de Rham moduli space.

Definition 7. Fix a smooth projective curve C of genus $g \geq 1$. Then we define $\mathcal{M}_{n,\mathrm{dR}}$ to parameterize (isomorphism classes of) pairs $(\mathcal{E}, D)$, where $\mathcal{E}$ is a smooth vector bundle on $C(\mathbb{C})$ and D is a flat connection on $\mathcal{E}$, and we require $(\mathcal{E}, D)$ is suitably semisimple.

Remark 8. The "isomorphsim classes" phrase means that we are considering the pairs up to gauge transformations.

One can get the same space $\mathcal{M}_{n,\mathrm{dR}}$ if we require that $\mathcal{E}$ be holomorphic and D to be flat holomorphic.

Theorem 9. Fix a smooth projective curve C of genus $g \geq 1$. Then a holomorphic vector bundle $\mathcal{E}$ on $C(\mathbb{C})$ has equivalent data to a smooth vector bundle $\mathcal{E}$ along with the data of a "holomorphic structure"

$$\overline{\partial}_{\mathcal{E}} \colon \mathcal{A}^0(\mathcal{E}) \to \mathcal{A}^{0,1}(\mathcal{E}).$$

Sketch. We need to explain what a holomorphic section of $\mathcal{E}$ is, for which we will take just to be the smooth sections which vanish under $\overline{\partial}_{\mathcal{E}}$. Now, given D, one can decompose $\mathcal{A}^1(X)$, so one can decompose $D = D^{0,1} + D^{1,0}$, which we label as $\overline{\partial}_{\mathcal{E}}$ and ∇. Then $D^2 = 0$ under the decomposition is equivalent to having $\overline{\partial}_{\mathcal{E}}^2 = 0$ (which gives the holomorphic structure on $\mathcal{E}$) and $\overline{\partial}_{\mathcal{E}}\nabla + \nabla\overline{\partial}_{\mathcal{E}} = 0$ (which means ∇ is holomorphic) and $\nabla^2 = 0$ (which means ∇ is flat). ■

We can now relate these two moduli spaces.

Theorem 10 (Riemann–Hilbert)**.** Fix a smooth projective curve C of genus $g \geq 1$. Then $\mathcal{M}_{n,\mathrm{B}}$ and $\mathcal{M}_{n,\mathrm{dR}}$ are isomorphic as manifolds.

Proof. The forward map sends a representation V to the pair $(V \otimes_{\mathbb{C}} \mathcal{O}_C, \nabla)$, where ∇ is some induced connection. The inverse map is given by holonomy theory. ■

1.3 Nonabelian Hodge Theory

Lastly, we need to tell a Dolbeault story. We will build a moduli space of Higgs bundles.

Definition 11 (Higgs bundle)**.** Fix a smooth projective curve C of genus $g \geq 1$. Then a *Higgs bundle* is a pair $(\mathcal{E}, \varphi)$, where $\mathcal{E}$ is a holomorphic vector bundle on C and $\varphi \colon \mathcal{E} \to \mathcal{E} \otimes K(C)$ is a Higgs field (meaning that it is $\mathcal{O}_C$-linear).

Remark 12. In general, one asks for $\varphi \wedge \varphi = 0$, but we do not need it here because we are over a cuve.

In order to get a moduli space, we need an adjective.

Definition 13 (semistable)**.** Fix a smooth projective curve C of genus $g \geq 1$. Then a Higgs bundle $(\mathcal{E}, \varphi)$ is *semistable* if and only if any nonzero proper φ-invariant sub-bundle $\mathcal{E}^1 \subseteq \mathcal{E}$ has $\mu\left(\mathcal{E}^1\right) \leq \mu(\mathcal{E})$. Here, $\mu(\mathcal{E}) \coloneqq \deg \mathcal{E} / \operatorname{rank} \mathcal{E}$.

Definition 14. Fix a smooth projective curve C of genus $g \geq 1$. Then we define $\mathcal{M}_{n,\mathrm{Dol}}$ to parameterize pairs $(\mathcal{E}, \varphi)$ of semistable Higgs bundles on C which satisfy $\deg \mathcal{E} = 0$.

Remark 15. If we want to replace C by a higher-dimensional variety, one should instead require that $c_i(\mathcal{E})_{\mathbb{Q}} = 0$ for all i instead of $\deg \mathcal{E} = 0$. (The former condition is stronger.)

Remark 16. Let's explain where such Higgs fields come from. Let $\mathrm{Bun}_n^0(C)$ be the moduli space of bundles on C of rank n and degree zero. For $[\mathcal{E}] \in \mathrm{Bun}_n^0(C)$, the tangent space at $[\mathcal{E}]$ should be given by deformations of $\mathcal{E}$, which should be given by $\mathrm{Ext}^1(\mathcal{E}, \mathcal{E})$, which by some duality is the same as $\mathrm{Hom}(\mathcal{E}, \mathcal{E} \otimes K(C))^\vee$. Thus, intuitively, $\mathcal{M}_{n,\mathrm{Dol}}$ should be the cotangent bundle of $\mathrm{Bun}_n^0(C)$.

Example 17. If $n = 1$, then $\mathcal{E}$ is a line bundle. A Higgs field $\varphi\colon \mathcal{L} \to \mathcal{L} \otimes K(C)$ has equivalent data to a global section of $K(C)$. Thus,

$$\mathcal{M}_{1,\mathrm{Dol}} \cong \mathrm{Pic}^0(C) \times \mathrm{H}^0(C; K(C)) \cong \mathbb{C}^g/\Lambda \otimes \mathbb{C}^g.$$

Thus, we see that $\mathcal{M}_{1,\mathrm{Dol}}$ and $\mathcal{M}_{1,\mathrm{B}}$ are isomorphic as manifolds but not as complex manifolds.

Theorem 18 (Nonabelian Hodge)**.** Fix a smooth projective curve C of genus $g \geq 1$. Then $\mathcal{M}_{n,\mathrm{dR}}$ and $\mathcal{M}_{n,\mathrm{Dol}}$ are isomorphic as manifolds.

Proof. The idea is to pass through some harmonic bundles. ∎

This is only a degree-zero story, so let's do some twisting to get higher degrees.

Definition 19. Fix a smooth projective curve C of genus $g \geq 1$, and choose an integer d coprime to n. Then we define the *twisted Betti moduli space* $\mathcal{M}^d_{n,\mathrm{B}}$ to be

$$\left\{(A_1, \dots, A_g, B_1, \dots, B_g) \in \mathrm{GL}_n(\mathbb{C}) : \prod_{i=1}^{g} [A_i, B_i] = e^{2\pi i d/n}\right\} / \mathrm{GL}_n(\mathbb{C}).$$

Remark 20. This admits a moduli description as local systems on the punctured curve with a specific monodromy at the puncture.

Definition 21. Fix a smooth projective curve C of genus $g \geq 1$, and choose an integer d coprime to n. Then we define $\mathcal{M}^d_{n,\mathrm{Dol}}$ to parameterize pairs $(\mathcal{E}, \varphi)$ of semistable Higgs bundles on C which satisfy $\deg \mathcal{E} = d$.

Theorem 22 (Twisted nonabelian Hodge)**.** Fix a smooth projective curve C of genus $g \geq 1$, and choose an integer d coprime to n. Then $\mathcal{M}_{n,\mathrm{dR}}$ and $\mathcal{M}_{n,\mathrm{Dol}}$ are isomorphic as manifolds.

Remark 23. The condition $\gcd(d, n) = 1$ is only needed to ensure smoothness everywhere.

1.4 The Statement

We are now in a position to define P and W. Let's start with W.

Definition 24 (Hodge structure)**.** A pure Hodge structure of weight $k \in \mathbb{Z}$ is a rational vector space H along with a decomposition

$$H_{\mathbb{C}} \cong \bigoplus_{p+q=k} H^{p,q}$$

for which $H^{p,q} = \overline{H^{q,p}}$.

Example 25. If X is a smooth projective complex variety, then $\mathrm{H}^k(X; \mathbb{C})$ is a pure Hodge structure of weight k. The Hodge structure is concentrated in pairs (p, q) where p and q are nonnegative.

One would like to port over this decomposition to the nonabelian setting.

Theorem 26 (Deligne)**.** Fix an algebraic variety X over $\mathbb{C}$. Then there is a canonical ascending weight filtration on $\left\{W_i \mathrm{H}^k(X;\mathbb{Q})\right\}_{0\le i\le 2k}$ such that

$$\frac{W_j \mathrm{H}^k(X;\mathbb{Q})}{W_{j-1}\mathrm{H}^k(X;\mathbb{Q})}$$

is a pure Hodge structure of weight j.

Applied to the affine algebraic variety $\mathcal{M}^d_{n,\mathrm{B}}$ gives rise to a weight filtration on its cohomology.

We now move on to P.

Definition 27 (Hitchin fibration)**.** Fix a Higgs bundle $(\mathcal{E},\varphi)$ by embedding $\mathcal{O}_C \subseteq K(C)$, we have a characteristic polynomial $\mathrm{charpoly}(\varphi) \coloneqq \det(T-\varphi)$. Then the *Hitchin map* is the associated map

$$h\colon \mathcal{M}^d_{n,\mathrm{Dol}} \to \bigoplus_{i=1}^{n} \mathrm{H}^0\left(C; K(C)^{\otimes i}\right)$$

sending $(\mathcal{E},\varphi)$ to the coefficients of the characteristic polynomial.

Theorem 28. The Hitchin map is proper, and the general fiber of h is an abelian variety and Lagrangian.

Remark 29. The symplectic structure on $\mathcal{M}^d_{n,\mathrm{Dol}}$ arises by viewing it as a cotangent bundle.

Example 30. If $n=1$, then the Hitchin map is the projection $\mathrm{Pic}^d(C)\times\mathbb{C}^g \to \mathbb{C}^g$.

One would like to realize the Hodge decomposition on the cohomology of $\mathcal{M}_{n,\mathrm{Dol}}$ as living over the Hitchin base. Well, set $K \coloneqq Rh_*\mathbb{Q}[2d]$, where d is the dimension of the base, and then a large theorem decomposes K into a perverse filtration.

The main goal of these talks is to prove the following theorem.

Theorem 31. The perverse and weight filtrations on the cohomology of $\mathcal{M}^d_{n,\mathrm{Dol}}$ and $\mathcal{M}^d_{n,\mathrm{B}}$ agree.

The idea will be to separately show that P and W both agree with a third filtration named C.

2 September 28: Hitchin Fibrations and Spectral Curves

This talk was given by Kaiyi at MIT for the $P=W$ seminar.

2.1 The Hitchin Fibration

Throughout, X is a smooth projective complex curve off genus g, where $g\ge 2$; set $\mathcal{K}_X$ to be the canonical bundle. Recall the following definition from last time.

Definition 32 (Higgs bundle)**.** A *Higgs bundle* is a pair $(\mathcal{E},\varphi)$ of a vector bundle $\mathcal{E}$ on X and a morphism $\varphi\in \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{E},\mathcal{E}\otimes\mathcal{K}_X)$.

Roughly speaking, $(\mathcal{E},\varphi)$ is a cotangent vector of the moduli space of vector bundles on X. For this talk, we are living in the Dolbeault setting, so we simply define $\mathcal{M}$ to be the moduli space of semistable Higgs bundles on rank n and degree zero.

We also constructed the Hitchin fibration.

Definition 33 (Hitchin fibration)**.** The *Hitchin base* $\mathcal{A}$ is

$$\mathcal{A} := \bigoplus_{i=1}^{n} \mathrm{H}^0(X; \mathcal{K}_X^{\otimes i}).$$

We now define the Hitchin map $f\colon \mathcal{M} \to \mathcal{A}$ to send $(\mathcal{E}, \varphi)$ to the characteristic polynomial.

Remark 34. On fibers, we are basically realizing φ as a matrix (upon choosing a trivialization of $\mathcal{K}_X$) and taking traces of its exterior powers. In a more coordinate-free way, one may think about the trace of φ as the composite

$$\mathcal{O}_X \to \mathcal{E} \otimes \mathcal{E}^\vee \xrightarrow{\varphi} \mathcal{E} \otimes \mathcal{K}_X \otimes \mathcal{E}^\vee \to \mathcal{K}_X.$$

Remark 35. Note $\dim \mathcal{M} = 2 \dim \mathcal{A}$. Indeed, $\dim \mathcal{M} = 2(\dim \mathrm{Bun}_G(X) + 1)$, where the $+1$ arises for stacky reasons, and $\dim \mathrm{Bun}_G(X) = n^2(g-1)$ by a tangent complex calculation. On the other hand, one can compute $\dim \mathcal{A}$ as some Riemann–Roch calculation.

Here is our main result for today.

Theorem 36. The map $f\colon \mathcal{M} \to \mathcal{A}$ is proper, and its general fibers are abelian varieties.

Remark 37. Roughly speaking, this amounts to showing that f is a complete integrable system. Notably, all fibers are Lagrangians.

2.2 Spectral Curves

Let's restate the data of a Higgs bundle more geometrically.

Definition 38 (cotangent bundle)**.** Define $\mathcal{S}$ to be the cotangent bundle of X, which is the vector bundle $\operatorname{Spec} \operatorname{Sym}_{\mathcal{O}_X} \mathcal{K}_X^\vee$. Let $\pi\colon \mathcal{S} \to X$ be the natural map.

Remark 39. A Higgs bundlehas equivalent data to a vector bundle $\mathcal{E}$ with a map $\mathcal{E} \otimes \mathcal{K}_X^\vee \to \mathcal{E}$, which is equivalent to the data of an action of $\operatorname{Sym}_{\mathcal{O}_X} \mathcal{K}_X^\vee$ on $\mathcal{E}$. (This is an enhancement of the fact that a vector bundle equipped with an endomorphism is equivalent to a $\mathbb{C}[T]$-module.) This turns out to be equivalent to the data of a quasicoherent sheaf $\mathcal{F}$ on $\mathcal{S}$ for which $\pi_* \mathcal{F}$ is actually a vector bundle on X.

This remark roughly proves the following proposition.

Proposition 40. The category of Higgs bundles of rank n is equivalent to the category of coherent sheaves $\mathcal{F}$ on $\mathcal{S}$ such that $\operatorname{supp} \mathcal{F} \to X$ is finite, and $\pi_* \mathcal{F}$ is torsion-free of rank n.

Proof. Note that torsion-free is equivalent to being a vector bundle. Finiteness of the support map should be formal. ∎

The support of the relevant $\mathcal{F}$ is our spectral curve.

Definition 41. Fix $(a_1, \dots, a_n) \in \mathcal{A}$. Then the *spectral curve* Y_a is the curve in $\mathcal{S}$ cut out by

$$T^n - \pi^* a_1 T^{n-1} + \cdots + (-1)^n \pi^* a_n.$$

Remark 42. By definition, the canonical map $\pi_a \colon Y_a \to X$ is finite of degree n.

Example 43. If $f(\mathcal{E}, \varphi) = a$, then the support of the relevant coherent sheaf $\mathcal{F}$ is set-theoretically the same as Y_a. (Due to reducedness issues, one has $\operatorname{supp} \mathcal{F} \subseteq Y_a$ in general.) One sees this by the construction of $\mathcal{F}$ via the induced action of $\operatorname{Sym}_{\mathcal{O}_X} \mathcal{K}_X^\vee$.

Let's check smoothness.

Proposition 44. For a general point $a \in \mathcal{A}$, the curve Y_a is smooth and connected.

Proof. Let's start with smoothness. One approach is to check this for a single a because this is an open condition. Alternatively, note that $\{Y_a\}_a$ is a linear system on $\mathcal{S}$, and the linear system has no basepoint: away from the zero section, one can always do some scaling to move points arounds, and on the zero section, we are basically asserting that $\mathcal{K}_X^{\otimes n}$ is base-point free. Bertini's theorem now gives the general smoothness.

We now move on to connectivity. Because Y_a is cut out by a monic polynomial of degree n, one finds that

$$\pi_* \mathcal{O}_{Y_a} = \mathcal{O}_X \oplus \mathcal{K}_X^{-1} \oplus \mathcal{K}_X^{-2} \oplus \cdots \oplus \mathcal{K}_X^{1-n}.$$

This implies that $\mathrm{H}^0(Y_a; \mathcal{O}_{Y_a})$ is one-dimensional. (Here we are using that $g \geq 2$.) ∎

It will turn out that the Jacobians of the spectral curves are the fibers of the Hitchin map.

Proposition 45. Suppose $a \in \mathcal{A}$ is chosen so that Y_a is smooth and reduced. Then $f^{-1}(\{a\})$ is a torsor for $\operatorname{Jac} Y_a$.

Proof. Suppose Y_a is reduced. Then $f^{-1}(a)$ consists of those Higgs bundles $(\mathcal{E}, \varphi)$ with spectral curve Y_a, which is in bijecetion with those coherent sheaves $\mathcal{F}$ on Y_a for which $\pi_* \mathcal{F}$ is a rank-n vector bundle. Because Y_a is reduced, this turns out to be the same as requiring that $\mathcal{F}$ is a coherent sheaf which is torsion-free and generically rank one, which is in almost parameterized by the compactified Jacobian. It turns out that $\deg_{Y_a} \mathcal{F} = \deg \mathcal{E} + n(n-1)(g-1)$ by some exercise in algebraic geometry, so we are actually looking at a torsor for $\operatorname{Jac} Y_a$. ∎

Remark 46. Even if not smooth, one can replace Jacobians with compatified Jacobians, at least for nodal curves.

This shows that the general fibers of f are abelian varieties.

2.3 Properness

It remains to discuss the properness of $f \colon \mathcal{M} \to \mathcal{A}$. It will be helpful to compactify everything in sight. Let $\mathcal{P}$ be the projectivization of $\mathcal{S}$, which is $\mathbb{P}(\mathcal{K} \oplus \mathcal{O}_X)$; the complement of S is D, and it is isomorphic to X (as the zero section). On the other side, let $\mathcal{M}_{\mathcal{P}}$ be the moduli space of coherent semistable sheaves on $\mathcal{P}$, which turns out to be projective.

Now, for $\mathcal{F} \in \mathcal{M}_{\mathcal{P}}$, note that $\mathcal{F} \in \mathcal{M}$ if and only if $\pi_* \mathcal{F}$ is a vector bundle of rank n, which is equivalent to its support avoiding D and $\pi_* \mathcal{F}$ being torsion-free of rank n. The condition of being torsion-free of rank n can be seen on the level of the Hilbert polynomial, so it is seen on the connected components of $\mathcal{M}_{\mathcal{P}}$ already. Thus, we will ignore this latter condition.

Remark 47. Observe that having support avoiding D is an open condition, so $\mathcal{M}$ itself is found as an open subset of $\mathcal{M}_{\mathcal{P}}$.

We will check the properness of f via the valuative criterion. Thus, we choose a diagram as follows, where R is a discrete valuation ring, and K is its fraction field.

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & \mathcal{M} \\ \downarrow & & \downarrow \\ \operatorname{Spec} R & \longrightarrow & \mathcal{A} \end{array}$$

We need to produce a lift from $\operatorname{Spec} R$ to $\mathcal{M}$. Because $\mathcal{M}_{\mathcal{P}}$ is already proper, we may lift our map to $\operatorname{Spec} R \to \mathcal{M}_{\mathcal{P}}$ (on a suitable compactification). Thus, we basically have a family of coherent sheaves $\mathcal{F}$ on $\mathcal{P}_R$, and we are given that their support on the fiber $\mathcal{P}_K$ lives over $Y_a \times \operatorname{Spec} K$. But taking the limit forces our support of $\mathcal{F}$ to continue to live over Y_a, which gives the correct lift.

We conclude the talk by saying something about Jacobians of nodal curves.

Proposition 48. Fix a nodal curve C with node c, and let $\nu\colon \widetilde{C} \to C$ be the normalization. Then $\overline{\operatorname{Pic} C} = \operatorname{Pic} C \sqcup \operatorname{Pic} \widetilde{C}$.

Proof. Recall that a nodal curve looks locally (at the node) like the covering $\mathbb{C}[[x,y]]/(xy) \subseteq \mathbb{C}[[x]] \times \mathbb{C}[[y]]$. Then this basically follows from the following piece of commutative algebra: there are exactly two torsion-free finitely generated generically rank-one modules over $\mathbb{C}[[x,y]]/(xy)$. Namely, one has the trivial bundle and the pushforward of $\mathbb{C}[[x]] \times \mathbb{C}[[y]]$. ■

Remark 49. The natural map $\operatorname{Pic} \widetilde{C} \to \overline{\operatorname{Pic} C}$ increases the degree by one.

Remark 50. Recall that $\operatorname{Pic}^0 C$ lives in the short exact sequence

$$0 \to \mathbb{G}_m \to \operatorname{Pic}^0 C \to \operatorname{Pic}^0 \widetilde{C} \to 0,$$

where $\mathbb{G}_m$ is the data of identifying two of the fibers. Thus, $\operatorname{Pic}^0 C$ is a semi-abelian variety.

Remark 51. Let's describe the scheme structure on $\overline{\operatorname{Pic}}^0(C)$. Fix two line bundles $\mathcal{L}_+$ and $\mathcal{L}_-$ on $\widetilde{C}$ which restrict to the same line L over the two points in c^+ and c^- in $\nu^{-1}(\{c\})$. Then set $Q \coloneqq \mathbb{P}^1(\mathcal{L}_+ \oplus \mathcal{L}_-)$, which admits a natural map to $\overline{\operatorname{Pic}}^0(C)$ by sending the quotient $\mathcal{L}_+ \oplus \mathcal{L}_- \to \mathcal{Q}$ to the kernel of $\nu_* L \to L|_{c^+} \oplus L|_{c^-}$. This turns out to normalize $\overline{\operatorname{Pic}}^0(C)$.

3 October 19: Mixed Hodge Structures

This talk was given by Merrick at MIT for the $P = W$ seminar.

3.1 Pure Hodge Structures

Let's start with pure Hodge structures.

Definition 52. A *pure Hodge structure* of weight $n \in \mathbb{Z}$ is a finite-dimensional $\mathbb{Q}$-vector space $V_{\mathbb{Q}}$ equipped with a decreasing filtration $\{F^p V_{\mathbb{C}}\}_p$ such that

$$V_{\mathbb{C}} = F^p V_{\mathbb{C}} + \overline{F^{n-p+1} V_{\mathbb{C}}}$$

for each p.

Remark 53. Given this filtration, one can set $V_{\mathbb{C}}^{p,q} \coloneqq F^pV \cap \overline{F^qV}$ to recover the usual Hodge decomposition.

Example 54. Fix a smooth projective variety X. Then the cohomology group $\mathrm{H}^n(X;\mathbb{Q})$ admits a natural structure of a pure Hodge structure of weight n.

Example 55 (Tate twist)**.** Fix an integer $i \in \mathbb{Z}$. Then we define the *Tate twist* $\mathbb{Q}(i)$ to be the pure Hodge structure of weight $-2i$ which is given by the $\mathbb{Q}$-vector space $V_{\mathbb{Q}} = \mathbb{Q}$ and Hodge decomposition supported at $V_{\mathbb{C}}^{-i,-i}$. The Hodge filtration is given by $F^{-i}V_{\mathbb{C}} = \mathbb{C}$ and $F^{-i+1}V_{\mathbb{C}} = 0$.

Now that we have objects, we want some morphisms.

Definition 56. A *morphism* of pure Hodge structures is a linear map $f_{\mathbb{Q}} \colon V_{\mathbb{Q}} \to V'_{\mathbb{Q}}$ of the underlying $\mathbb{Q}$-vector spaces which preserves the filtrations. A *strict morphism* is one for which $\operatorname{im} f_{\mathbb{C}} \cap F^pV'_{\mathbb{C}} = f_{\mathbb{C}}(F^pV_{\mathbb{C}})$ for all p.

Remark 57. If $V_{\mathbb{Q}}$ and $V'_{\mathbb{Q}}$ have the same weight, then any morphism is strict.

Remark 58. The category of pure Hodge structures of fixed weight is abelian.

Example 59. It turns out that $\mathrm{H}^0(\mathrm{pt};\mathbb{Q}) \cong \mathbb{Q}(0)$. More generally, $\mathrm{H}^{2r}(\mathbb{P}^r;\mathbb{Q}) \cong \mathbb{Q}(-r)$.

Example 60. It turns out that $\mathrm{H}^1(\mathbb{G}_m;\mathbb{Q}) \cong \mathbb{Q}(-1)$.

Example 61. Let C be a smooth projective curve of genus g, and consider the pure Hodge structure $\mathrm{H}^1(C;\mathbb{Q})$. Then the Hodge filtration is given by $\mathrm{H}^{1,0} = \mathrm{H}^0\left(C;\Omega^1\right)$, which is $F^1\mathrm{H}^1(C;\mathbb{C})$.

3.2 Mixed Hodge Structures

We now pass to mixed Hodge structures.

Definition 62 (mixed Hodge structure)**.** A *mixed Hodge structure* V is a finite-dimensional $\mathbb{Q}$-vector space $V_{\mathbb{Q}}$ along with an increasing *weight filtration* $\{W_\ell V_{\mathbb{Q}}\}_\ell$ and a decreasing *Hodge filtration* $\{F^pV_{\mathbb{C}}\}_p$ such that the ℓth subquotient $\operatorname{Gr}_\ell V_{\mathbb{Q}}$ is a pure Hodge structure of weight ℓ. There is a natural notion of morphism.

Example 63. Any pure Hodge structure is a mixed Hodge structure.

Remark 64. Any morphism of mixed Hodge structure is suitably strict automatically, with respect to both filtrations. For example, this implies that there are no morphisms $V \to W$ between two Hodge structures V and W if the weights of V are supported above $w+1$ and the weights of W are below w.

Remark 65. From a mixed Hodge structure V, one can (as with pure Hodge structures), define $I^{p,q} \subseteq V_{\mathbb{C}}$ to satisfy $W_\ell V = \bigoplus_{p+q\leq \ell} I^{p,q}$ and $F^p V = \bigoplus_{r\geq p}\bigoplus_q I^{r,q}$.

Remark 66. A map of mixed Hodge structures which is an isomorphism on the underlying $\mathbb{Q}$-vector spaces is an isomorphism of the mixed Hodge structures.

Remark 67. The category of mixed Hodge structures is abelian. It is also symmetric monoidal, where the product is given by the usual tensor product.

Example 68. Define $V_{\mathbb{Q}} \coloneqq \mathbb{Q}\{a,b\}$. We give $V_{\mathbb{Q}}$ a weight filtration by $W_0V = W_1V = \mathbb{Q}\{a\}$ and $W_2V = V_{\mathbb{Q}}$, and we give $V_{\mathbb{C}}$ a Hodge filtration by $F^0V = V_{\mathbb{C}}$ and $F^1V_{\mathbb{C}} = \mathbb{C}\{b+ai\}$ and $F^2V_{\mathbb{C}} = 0$. This forms a mixed Hodge structure. For example, $\mathrm{Gr}_0^W V = \mathbb{Q}\{a\}$, which is a pure Hodge structure of type $(0,0)$. Also, $\mathrm{Gr}_1^W V = 0$ and $\mathrm{Gr}_2^W V = \mathbb{Q}\{a,b\}/\mathbb{Q}\{a\}$, which is then concentrated in F^1V so that it is a pure Hodge structure of type $(1,1)$. It turns out that V is not a direct sum of pure Hodge structures: it gives a non-split extension

$$0 \to \mathbb{Q}(0) \to V \to \mathbb{Q}(-1) \to 0.$$

In short, a splitting would mean that V is isomorphic to $\mathbb{Q}(0)\oplus\mathbb{Q}(-1)$, but $\mathbb{Q}(-1)$ would then need to arise as $\mathbb{Q}\{b+qa\}$ for some rational q.

3.3 Mixed Hodge Complexes

We would now like to do something derived.

Definition 69. A *mixed Hodge complex* $A^\bullet$ is a pair of cochain complexes $A^\bullet_{\mathbb{Q}} \in \mathrm{CoCh}(\mathrm{Vec}_{\mathbb{Q}})$ and $A^\bullet_{\mathbb{C}} \in \mathrm{CoCh}(\mathrm{Vec}_{\mathbb{C}})$, a weight filtration $\widetilde{W}_\ell A^\bullet_{\mathbb{Q}}$ on the complex, a Hodge filtration $\widetilde{F}^p A^\bullet_{\mathbb{C}}$, and a filtred quasi-isomorphism $A^\bullet_{\mathbb{Q}}\otimes\mathbb{C} \to A^\bullet_{\mathbb{C}}$. This data is required to satisfy the following.

- The graded differential $\mathrm{Gr}_\ell^{\widetilde{W}} A^\bullet_{\mathbb{C}} \to \mathrm{Gr}_\ell^{\widetilde{W}} A^{\bullet+1}_{\mathbb{C}}$ is strictly compatible with respect to the Hodge filtration $\widetilde{F}^p$.
- For all n, the cohomology $\mathrm{H}^n(\mathrm{Gr}_\ell^{\widetilde{W}} A^\bullet_{\mathbb{Q}})$ with the induced Hodge filtration is a pure Hodge structure of weight $n+\ell$.

Remark 70. The first point implies that cohomology commutes with $\mathrm{Gr}_\ell^{\widetilde{W}}$.

Remark 71. Importantly, $A^\bullet_{\mathbb{C}}$ is not $A^\bullet_{\mathbb{Q}}\otimes\mathbb{C}$ by definition. The following example explains why.

Example 72. Fix a smooth projective variety X. Then the singular cochain complex of X (with its weight filtration) together with the de Rham cochain complex of X (with its Hodge filtration) forms a mixed Hodge complex. Notably, $A^\bullet_{\mathbb{C}}$ and $A^\bullet_{\mathbb{Q}}\otimes\mathbb{C}$ are not obviously isomorphic, but the comparison theorem provides a filtered quasi-isomorphism.

Theorem 73. Fix a mixed Hodge complex $A^\bullet$.

(a) The spectral sequence induced by the weight filtration degenerates at E_2.

(b) The spectral sequence induced by the Hodge filtration degenerates at E_1.

(c) Taking cohomology induces a mixed Hodge structure: for each n, define

$$\begin{cases} W_\ell \mathrm{H}^n \coloneqq \operatorname{im}\left(\mathrm{H}^n(\widetilde{W}_{\ell-n}A^\bullet_{\mathbb{Q}}) \to \mathrm{H}^n(A^\bullet_{\mathbb{Q}})\right), \\ F^p\mathrm{H}^n \coloneqq \operatorname{im}\left(\mathrm{H}^n(\widetilde{F}^pA^\bullet_{\mathbb{C}}) \to \mathrm{H}^n(A^\bullet_{\mathbb{C}})\right). \end{cases}$$

Let's now begin a geometric example. Throughout, fix a smooth quasi-projective complex variety X, and suppose we have a smooth projective compactification $j\colon X \to \overline{X}$, where $D \coloneqq \overline{X} \setminus X$ is a normal crossings divisor.

Definition 74. Fix $j\colon X \to \overline{X}$ as above. Then we define $\Omega^1_{\overline{X}}(\log D)$ is freely generated over $\mathcal{O}_X$ near a point $p \in D$ where D is cut out as $z_1 \cdots z_r = 0$ by the differentials

$$\{dz_1/z_1, \dots, dz_r/z_r, dz_{r+1}, \dots, dz_{\dim X}\}.$$

We then define $\Omega^k_{\overline{X}}(\log D) \coloneqq \wedge^k \Omega^1_{\overline{X}}(\log D)$. Intuitively, we are permitting logarithmic singularities.

Remark 75. These constructions turn out to not depend on the choice of local coordinates $\{z_\bullet\}$

Example 76. It turns out that $\Omega^\bullet_{\overline{X}}(\log D) \simeq \mathrm{R}^\bullet j_*\mathbb{C}_X$. This gives a mixed Hodge complex (of sheaves).

- One gets a weight filtration on $A^\bullet_{\mathbb{Q}} \coloneqq \mathrm{R}^\bullet j_*\mathbb{Q}_X$ by truncation as $\widetilde{W}_m A^\bullet_{\mathbb{Q}} \coloneqq \tau_{\le m}\mathrm{R}^\bullet j_*\mathbb{Q}_X$.
- One gets a Hodge filtration on $A^\bullet_{\mathbb{C}} \coloneqq \Omega^\bullet_{\overline{X}}(\log D)$ by $\widetilde{F}^p \coloneqq \Omega^{\ge p}_{\overline{X}}(\log D)$.

There is also a weight filtration concentrated in nonnegative degrees with $\widetilde{W}_m A^\bullet A^k = A^k$ when $k \ge m$ and $\widetilde{W}_m A^k = \Omega^{k-m}_{\overline{X}} \wedge \Omega^m_{\overline{X}}(\log D)$ when $m \le k$. It turns out that this weight filtration recovers the original weight filtration via the quasi-isomorphism.

Example 77. It turns out that the moduli space of Higgs bundles has cohomology of pure weight.

Example 78. Consider the cohomology of $\mathbb{G}_m$. It turns out that the weight filtration is concentrated in degree two. Indeed, there is a short exact sequence

$$0 \to i^!\mathbb{Q}_{\mathbb{P}^1} \to \mathbb{Q}_{\mathbb{P}^1} \to j_*\mathbb{Q}_{\mathbb{G}_m} \to 0.$$

It turns out that $i^! = i^*[2](1)$, and taking cohomology allows one to compute some weights by truncating.

3.4 Simplicial Varieties

We close our talk by saying something about simplicial varieties. Fix a simplicial variety $X_\bullet$, which is basically a functor $\Delta^{\mathrm{op}} \to \mathrm{Var}_{\mathbb{C}}$. Such a simplicial variety $X_\bullet$ has a topological realization $|X_\bullet|$ (constructed after taking complex points).

Definition 79. Fix a simplicial variety $X_\bullet$. Then an *augmentation* is a map $\varepsilon\colon X_0 \to Y$ for which $\varepsilon \circ d_0 = \varepsilon \circ d_1$, where $d_0, d_1\colon X_1 \to X_0$ are the projections. We may call ε an *augmented* simplicial variety, denoted $\varepsilon\colon X_\bullet \to Y$.

Remark 80. An augmentation induces a map on the topological realizations.

Definition 81 (hypercover)**.** An augmented simplicial variety $\varepsilon\colon X_\bullet \to X$ is a *hypercovering* if and only if $|\varepsilon| : |X_\bullet| \to X$ has contractible fibers.

Example 82. Our key example comes from a compactification $j\colon X \to \overline{X}$. One takes X_0 to be the compactification $\overline{X}$, we take X_1 to be the divisors present in $D \coloneqq \overline{X} \setminus X$, we take to be X_2 to be the intersections of the divisors in D, and so on.

Remark 83. If X is singular, then some process of resolution of singularities allows one to construct a simplicial smooth hypercovering $X_\bullet \to X$. If X is proper, then one may make the simplicial hypercovering compact.

Simplicial varieties provide examples of mixed Hodge complexes.

Notation 84. Fix a simplicial variety $X_\bullet$. Then we define

$$K_{\mathbb{Q}}^n \coloneqq \bigoplus_{p+q=n} C_{\mathrm{sm}}^q(X_p;\mathbb{Q}) \qquad \text{and} \qquad K_{\mathbb{C}}^n \coloneqq \bigoplus_{p+q=n} A^q(X_p).$$

Here, the differential is induced by taking the total complex of the associated bi-complexes. Note that these are quasi-isomorphic after base-changing to $\mathbb{C}$.

Remark 85. There is a Hodge filtration $\widetilde{F}$ by defining $\widetilde{F}^a K_{\mathbb{C}}^n$ as usual by $\bigoplus_{p+r+s=n, r\geq a} A^{r,s}(X_p)$. There is a weight filtration $\widetilde{W}$ by truncating to $\widetilde{W}_m K_{\mathbb{Q}}^n \coloneqq \bigoplus_{p+q=n, p\geq -m} C_{\mathrm{sm}}^q(X_p;\mathbb{Q})$.

The moral is that we can get a mixed Hodge structure on the cohomology of X, even when X fails to be smooth or proper.

4 October 26: Perverse Sheaves and Perverse Filtrations

This talk was given by Yutong at MIT for the $P = W$ seminar.

4.1 Constructible Sheaves

Today, X is a complex variety, and our sheaves are valued in vector spaces over $\mathbb{Q}$. Throughout, $\mathrm{Sh}(X)$ is the category of sheaves, and $D^b(X) \coloneqq D^b(\mathrm{Sh}(X))$.

Definition 86 (local system)**.** A *local system* is a locally constant sheaf of rational vector spaces.

Definition 87 (stratification)**.** Fix a complex variety X. Then a *stratification* is a collection of "smooth strata" $\{X_s\}_{s\in S}$ covering X such that each X_s is locally closed, and each $\overline{X}_s$ is a union of strata. We will usually assume that S is finite. We may say that X is *stratified*.

Definition 88 (constructible)**.** Fix a stratified complex variety X. Then a sheaf $\mathcal{F}$ is *constructible* with respect to the stratification if and only if $\mathcal{F}|_{X_s}$ is a local system for each stratum X_s. If $\mathcal{F}$ is constructible with respect to some stratification, then we simply call $\mathcal{F}$ *constructible*. A complex is *constructible* if and only if its cohomology sheaves are constructible; we denote this category by $D^b_c(X)$ and $D^b_S(X)$, where the latter requires stratification by S.

This category has six functors.

Remark 89. The usual pullback functor is exact, and it restricts to complexes of constructible sheaves.

Remark 90. Fix $f\colon X \to Y$. The functor $f_*\colon \operatorname{Sh} X \to \operatorname{Sh} Y$ is left-exact, so it produces $\mathrm{R}f_*\colon D^b(X) \to D^b(Y)$, which restricts to constructible sheaves. We may abbreviate $\mathrm{R}f_*$ to f_*. Note that $\mathrm{R}f_*$ is right adjoint to f_*.

Remark 91. Fix $f\colon X \to Y$. There is a functor $f_!\colon \operatorname{Sh} X \to \operatorname{Sh} Y$ given by

$$f_!\mathcal{F}(U) \coloneqq \{s \in f_*\mathcal{F}(U) : f|_{\operatorname{supp} s}\colon \operatorname{supp} s \to U \text{ is proper}\}.$$

This functor continues to be left exact and induces $\mathrm{R}f_!\colon D^b_c(X) \to D^b_c(Y)$. We may abbreviate $\mathrm{R}f_!$ to $f_!$.

Remark 92. Fix $f\colon X \to Y$. The functor $f_!$ has a right adjoint $f^!\colon D^b_c(Y) \to D^b_c(X)$. Here, $f^!$ can be constructed via some adjoint functor theorems; alternatively, here are two special cases.

- If f is smooth of relative dimension d, then $f^! \cong f^*[2d](d)$. Here, the Tate twist is given by $\mathbb{Q}(-1) \coloneqq \mathrm{H}^2_c\left(\mathbb{A}^1\right)$.
- If f is a locally closed embedding, there is another

Remark 93. There is an internal hom $\mathcal{H}om\colon \operatorname{Sh}(X)^{\mathrm{op}} \times \operatorname{Sh}(X) \to \operatorname{Sh} X$. It is left-exact in both variables, so we get $\mathrm{RHom}\colon D^b_c(X)^{\mathrm{op}} \times D^b_c(X) \to D^b_c(X)$.

Remark 94. Let $\pi\colon X \to \mathrm{pt}$ be the projection. Then there is a dualizing sheaf $\omega_X \coloneqq \pi^!\mathbb{C}$. For exapmple, if X is smooth, then $\omega_X = \mathbb{C}[2\dim X](\dim X)$. We may define the Verdier duality functor $\mathbb{D} \coloneqq \operatorname{Hom}(-, \omega_X)$, which is an involution $D^b_c(X)^{\mathrm{op}} \to D^b_c(X)$. It turns out that

Remark 95. The category $D^b_{\mathrm{con}}(X)$ is triangulated. The most basic example of a distinguished triangle is $U \subseteq X \supseteq Z$ where $i\colon U \to X$ is open and $j\colon Z \to X$ is its complement. Then any complex $\mathcal{F}$ admits the distinguished triangles

$$j_!j^*\mathcal{F} \to \mathcal{F} \to i_*i^*\mathcal{F} \xrightarrow{[1]} \qquad \text{and} \qquad i_*i^!\mathcal{F} \to \mathcal{F} \to j_*j^*\mathcal{F} \xrightarrow{[1]}.$$

4.2 Perverse Sheaves

There is a general machine which takes a t-structure on a triangulated category and outputs an abelian category.

Definition 96 (t-structure)**.** Fix a triangulated category $\mathcal{C}$. Then a *t-structure* is a triple $(\mathcal{T}^{\leq 0}, \mathcal{T}^{\geq 0}, [1])$ satisfying

- $\mathcal{T}^{\leq 0}$ and $\mathcal{T}^{\geq 0}$ are subcategories of $\mathcal{C}$,
- $[1]\colon \mathcal{C} \to \mathcal{C}$ is an auto-equivalence,
- the inclusion $\mathcal{T}^{\leq n} \coloneqq \tau^{\leq 0}[-n] \hookrightarrow \mathcal{C}$ admits a right adjoint $\tau^{\leq n}$, and
- the inclusion $\mathcal{T}^{\geq n} \coloneqq \mathcal{T}^{\geq 0}[-n] \hookrightarrow \mathcal{C}$ admits a left adjoint $\tau^{\geq n}$.

In this situation, the *heart* is the abelian catgeory $\mathcal{T}^{\geq 0} \cap \mathcal{T}^{\leq 0}$.

Definition 97. Define the *perverse t-structure* on $D_c^b(X)$ so that $[1]$ is the usual shift, $D_c^b(X)^{\leq 0}$ consists of those sheaves $\mathcal{F}$ for which $\dim \operatorname{supp} \mathrm{H}^i(\mathcal{F}) \leq -i$, and dually $D_c^b(X)^{\geq 0}$ is the image of the Verdier dual of $D_c^b(X)^{\leq 0}$. The category $\operatorname{Perv} X$ of *perverse sheaves* is the heart of the perverse t-structure.

Remark 98. If we fix a stratification $\{X_s\}_{s \in S}$, then pulling back our complex to each stratum reveals

$$D_S^b(X)^{\leq 0} = \{\mathcal{F} : i_s^* \mathcal{F} \text{ is concentrated in degrees } \leq -\dim X_s \text{ for all } s\},$$

and

$$D_S^b(X)^{\leq 0} = \{\mathcal{F} : i_s^* \mathcal{F} \text{ is concentrated in degrees } \geq -\dim X_s \text{ for all } s\}.$$

Example 99. Suppose X is smooth of dimension d. If $\mathcal{L}$ is a local system on X, then $\mathcal{L}[d]$ is a perverse sheaf.

Remark 100. Here is one source of the perverse t-structure: the Riemann–Hilbert correspondence implies that the (bounded) derived catgeory of holonomic D-modules on X with regular singularities is equivalent to $D_c^b(X)$. It turns out that the natural t-structure on $D^b(\mathrm{Hol}_{\mathrm{rs}}(X))$ corresponds to the perverse t-structure on $D_c^b(X)$.

4.3 Simple Perverse Sheaves

Here is an important source of such sheaves.

Definition 101 (intermediate extension)**.** Fix a locally closed embedding $h\colon Y \to X$. Then the *intermediate extension* is a functor $h_{!*}\colon \operatorname{Perv} Y \to \operatorname{Perv} X$ given by

$$h_{!*}\mathcal{F} \coloneqq \operatorname{im}\left({}^p\mathrm{H}^0(h_!\mathcal{F}) \to {}^p\mathrm{H}^0(h_*\mathcal{F})\right).$$

It turns out that this functor is fully faithful.

Example 102. If $\mathcal{L}$ is a local system on Y, then we define $\mathrm{IC}(\mathcal{L}) = h_{!*}(\mathcal{L}[\dim Y])$. We may also abbreviate $\mathrm{IC}_X \coloneqq \mathrm{IC}(\mathbb{Q}_X)$.

Proposition 103. Fix a locally closed embedding $h\colon Y \to X$. Then $h_{!*}$ is characterized by the following properties.

(a) For each $\mathcal{F}$, the complex $h_{!*}\mathcal{F}$ is supported on $\overline{Y}$.

(b) One has $h_{!*}\mathcal{F}|_Y \cong \mathcal{F}$.

(c) The complex $h_{!*}\mathcal{F}$ has no subobjects or quotients which are supported on $\overline{Y} \setminus Y$.

Proof. Here, (a) follows because $h_!\mathcal{F}$ and $h_*\mathcal{F}$ are both supported on $\overline{Y}$. We say nothing about the other two. ∎

The use of this construction is the following.

Proposition 104. The irreducible objects in $\mathrm{Perv}_S X$ are precisely those of the form $\mathrm{IC}(X_s, \mathcal{L})$, where $\mathcal{L}$ is a local system on X_s.

Proof. Omitted. ∎

Remark 105. The category $\mathrm{Perv}_S X$ is usually not semisimple.

Remark 106. It turns out that $\mathbb{D}(\mathrm{IC}(X_s, \mathcal{L})) = \mathrm{IC}(X_s, \mathcal{L}^\vee)$ for any line bundle $\mathcal{L}$ on a stratum X_s.

Let's focus on semisimple objects.

Definition 107 (semisimple)**.** A complex $\mathcal{F} \in D_b^c(X)$ is *semisimple* if and only if it is a direct sum of shifted simple perverse sheaves.

Remark 108. By splitting everything into shifted simple perverse sheaves, which are then IC-sheaves, we find that

$$\mathcal{F} \cong \bigoplus_n {}^p\mathrm{H}^n(\mathcal{F})[-n].$$

In particular, local systems are semisimple.

Theorem 109 (Decomposition)**.** Suppose $f\colon X \to Y$. Then f_* preserves semisimple objects.

Remark 110. If X and Y are both irreducible, then $f_* \mathrm{IC}_X$ contains $\mathrm{IC}_Y[\dim X - \dim Y]$.

4.4 The Perverse Filtration

Before being done for today, we should probably define the perverse filtration. Given a proper morphism $f\colon X \to Y$, we would like to undetstand $\mathrm{H}^*(X;\mathbb{Q}) = \mathrm{H}^*(Y; f_*\mathbb{Q})$.

Now, for any t-structure on $D_c^b(Y)$, there is a truncation $\tau^{\le n}\colon D_c^b(Y) \to D_c^b(Y)$, and we may define a filtration

$$\mathcal{F}_n\mathrm{H}^*(Y; f_*\mathbb{Q}) := \mathrm{im}\left(\mathrm{H}^*(Y; \tau^{\le n} f_*\mathbb{Q}) \to \mathrm{H}^*(Y; f_*\mathbb{Q})\right).$$

For example, using the classical t-structure produces the Leray filtration (found in the Leray spectral sequence). If we use the perverse t-structure, then we get a perverse filtration.

Remark 111. To compute the associated graded of this filtration, one can use the spectral sequence attached to this filtered complex. Using Theorem 109, one can show that this spectral sequence degenerates on E_2 (meaning that the perverse filtration has non-canonical splitting).

Let's give one more theorem.

Theorem 112. Suppose $f\colon X \to Y$ is proper with Y affine. Then there is an increasing filtration $\{Y_i\}_i$ of Y (and an inverse image flag in X) such that the perverse filtration on $\mathrm{H}^*(X;\mathcal{F})$ arises as a kernel from this filtration.

The moral is that the pieces (and associated graded) of the perverse filtration admits a mixed Hodge structure.